

BioLearnX: A Self-Improving Multi-Agent Framework for Personalized Biology Learning through Interactive Visualization, Adaptive Assessment, and Multilingual Content Generation

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Abstract—Biology learners typically move between separate tools for diagrams, revision, assessment, and note-taking, and few of these tools adapt to the individual learner or improve from experience. This paper presents BioLearnX, a multi-agent research framework that investigates whether four complementary agentic subsystems — interactive visual generation, adaptive assessment, multilingual audio review, and handwritten-note recovery — can share a common architectural pattern: retrieval-grounded large language model (LLM) agents, orchestrated as explicit graphs, that are evaluated by independent critics and improved only through validation-gated feedback. The visualization subsystem combines a catalog-driven anatomy specification extracted by an instruction model, image synthesis with a rectified-flow diffusion transformer, a dual CLIP/vision-language-model critic with a reflection-and-retry loop, segmentation-grounded interactive region explanation, and two-dimensional-to-3D asset conversion. The assessment subsystem generates curriculum-grounded, Bloom’s-taxonomy-mapped questions from learner-uploaded documents and adapts difficulty from response history. The note-recovery subsystem applies fine-tuned super-resolution and optical character recognition models to enhance and transcribe low-quality Sinhala handwriting, with Tamil and English translation. The audio-review subsystem provides multilingual (Tamil, Sinhala, English, and code-mixed) spoken question answering grounded in the learner’s own material. All four subsystems are implemented as independently deployable services on serverless GPU infrastructure and share a feedback-to-memory-to-skill-evolution lifecycle that gates every behavioral change behind held-out, paired evaluation. We describe the system design and the reproducible evaluation protocols implemented across the framework. An end-to-end baseline evaluation against the live deployment compares raw FLUX.1-dev output with our full visual pipeline: the proposed pipeline achieves substantial gains in VLM visual alignment (8.67 ± 0.58 vs. 6.00 ± 3.46 , +44.5%) and pedagogical usefulness (7.67 ± 0.58 vs. 5.00 ± 3.46 , +53.4%), consistently producing 8 verified, callout-annotated anatomical structures per image versus 0 for the unlabeled baseline, while collapsing generation variance ($\sigma=0.58$ vs. 3.46).

Keywords—multi-agent systems, large language models, retrieval-augmented generation, parameter-efficient fine-tuning, text-to-image generation, optical character recognition, adaptive learning, biology education.

I. INTRODUCTION

Learning biology well requires moving fluidly between several representations of the same material: a labeled diagram of an organ system, a self-test that adapts to what the learner has already mastered, a spoken explanation for revision on the move, and usable notes recovered from a lecture photograph or a handwritten page. Commercial and open-source tools typically address one of these modes in isolation, and few of them adapt their own behavior from evidence of what worked for a given learner or a given class of requests. Two distinct problems are

therefore left largely unaddressed by existing tools: (1) fragmentation across modalities — visual, assessment, spoken, and written — that a single learning session actually needs, and (2) the absence of a disciplined mechanism for an AI learning tool to improve itself without silently degrading correctness or safety.

Recent generative-AI research in anatomy and medical education has begun to explore embodied, conversational virtual assistants in immersive environments [7], and separate lines of work have produced strong building blocks that make a more integrated approach practical: instruction-tuned and vision-language foundation models that can be adapted cheaply with low-rank fine-tuning [1], [2]; promptable image segmentation that generalizes across domains [3]; rectified-flow diffusion transformers capable of photorealistic, instruction-following image synthesis [4]; and agent frameworks — reflection [5], self-refinement [6], and explicit stateful orchestration graphs — that let an LLM critique and retry its own output before it reaches a learner. What has not been demonstrated is a single research framework that applies these building blocks consistently across visualization, assessment, spoken review, and document recovery, under one shared, evidence-gated improvement lifecycle.

This paper presents **BioLearnX**, a four-subsystem research framework built to close that gap for biology education. Each subsystem is designed and implemented independently by one member of the project team, but all four share the same underlying principles: agentic pipelines orchestrated as explicit graphs rather than a single monolithic prompt; retrieval that grounds generation in the learner’s own curriculum material; an explicit separation between raw user feedback (never trusted directly) and reviewed, evidence-backed memory or skill updates; and independent, automated critics that gate output quality before a result reaches the learner.

The primary contributions of this paper are:

1. A catalog-driven, LoRA-adapted **interactive biology visualization pipeline** converting natural-language requests into validated anatomical specifications, synthesized graphics, critiqued outputs, interactive region explanations, and 3D assets (Section IV-A).
2. An **adaptive, curriculum-grounded assessment pipeline** generating Bloom’s-taxonomy-mapped questions from uploaded material and dynamically adapting difficulty (Section IV-B).
3. A **fine-tuned super-resolution and OCR pipeline** for low-quality Sinhala handwriting with Tamil/English translation

(Section IV-C).

4. A **multilingual, retrieval-grounded spoken question-answering pipeline** for conversational revision in Tamil, Sinhala, English, and code-mixed speech (Section IV-D).
5. A **shared, validation-gated feedback-to-skill-evolution lifecycle** ensuring behavioral updates deploy only after empirical verification (Section III-C).

II. RELATED WORK

A. Diffusion-Based and Agentic Image Generation

Rectified-flow diffusion transformers such as FLUX.1 [4] enable high-fidelity text-to-image synthesis, but unconditioned single-pass sampling provides no guarantees of biological correctness. Agentic loops address this: Reflexion [5] and Self-Refine [6] show that verbal reflection and critique significantly boost factual alignment. BioLearnX applies an explicit, independent dual critic rather than self-critique, decoupling evaluation from targeted regeneration.

B. Vision-Language Grounding and Segmentation

Interactive visual learning requires pixel-precise masks and grounded semantic reasoning. SAM 2 [3] generates high-quality segmentations without semantic bias, while Qwen2.5-VL [8] provides native coordinate-grid bounding boxes and pixel-verified visual question answering.

C. Parameter-Efficient Fine-Tuning

Full parameter fine-tuning is computationally impractical for specialized tasks. Low-Rank Adaptation (LoRA) [1] and QLoRA [2] enable rapid domain adaptation on single GPUs, applied here for structured anatomy specification extraction and Indic handwriting recognition.

D. Retrieval-Augmented Generation and Agent Memory

Sentence-BERT [9] and Hierarchical Navigable Small World (HNSW) index graphs [10] provide low-latency semantic retrieval. BioLearnX pairs dense retrieval with structured Memento memory, separating raw interactions from promoted skills.

III. SYSTEM ARCHITECTURE & SHARED DESIGN

BioLearnX is organized as a research monorepo of four independently deployable services sharing common design patterns (Fig. 1).

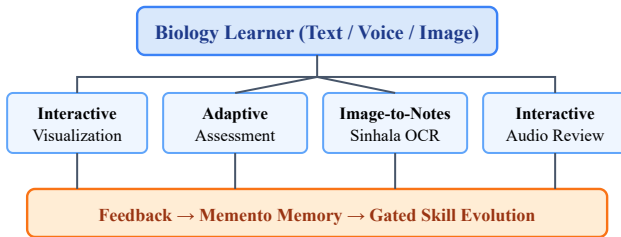


Fig. 1. BioLearnX system overview. Four independent agentic subsystems operate over shared feedback-to-skill evolution lifecycles.

A. Agentic Graph Orchestration

Pipelines are structured as explicit state graphs using LangGraph [11], executing over independent serverless Modal endpoints behind FastAPI interfaces. State errors are isolated per node without corrupting overall pipeline state.

B. Feedback-to-Skill Evolution Lifecycle

User feedback triggers targeted agent retries. Accepted revisions form immutable preference pairs. Repeated patterns synthesise a candidate skill document v' from the currently deployed v . The candidate becomes *eligible* only if it beats the deployed version by a margin δ on a held-out validation prompt set H , scored by the same dual critic that gates ordinary generation:

$$\Delta(v', v) = (1/|H|) \sum_{p \in H} [S_{\text{VLM}}(v', p) - S_{\text{VLM}}(v, p)] > \delta \quad (1)$$

Our deployment uses $\delta = 0.10$ with $|H| \geq 10$ held-out prompts, and the candidate is only synthesised once at least 50 high-scoring experiences have accumulated. Clearing (1) makes a version eligible, not live: deployment still requires explicit human approval, and the previous version is retained for rollback. Safety checks sit outside the editable skill document, so no evolution step can weaken them.

IV. METHODOLOGY: THE FOUR SUBSYSTEMS

A. Interactive Biology Visualization (EduVision)

Transforms natural language requests into explorable biological visuals through five coordinated agents:

1) *Prompt Agent*: Extracts canonical organ schemas (brain, heart, kidneys, liver, lungs) and structural constraints via fine-tuned Qwen2.5-3B-Instruct [12].

2) *Image Agent*: Generates high-resolution base images with FLUX.1-dev [4].

3) *Dual Critic & Reflection*: Evaluates CLIPScore [13] and VLM visual/pedagogical scores (Qwen2.5-VL [8]). Rejects embedded labels and triggers targeted prompt/seed modifications on failure.

4) *Interactive Agent*: Applies SAM 2 [3] and Qwen2.5-VL for grounded region segmentation, visual Q&A, and SVG leader-line annotations.

5) *3D Mesh Agent*: Converts approved 2D cutaways into textured 3D GLB meshes via Hunyuan3D-2 [14].

The dual critic scores every candidate image on two deliberately independent axes rather than one blended number. The reference-free alignment term follows CLIPScore [13], computed as the clamped cosine similarity between the CLIP image and text embeddings f_I and f_T of image I and its request T , on a 0–100 scale:

$$\text{CLIP}(I, T) = 100 \cdot \max(0, \langle f_I(I), f_T(T) \rangle / \|f_I(I)\| \|f_T(T)\|) \quad (2)$$

The semantic term is a vision-language judgment that returns a visual-alignment score S_{vis} and a pedagogical-utility score S_{ped} , reported both separately and as the composite

$$S_{\text{VLM}} = (S_{\text{vis}} + S_{\text{ped}}) / 2 \quad (3)$$

An image is released to the learner only when both axes clear the threshold $\tau = 7.0$ and the critic reports no hard failure ($F = 0$) — wrong laterality, a missing required structure, or text baked into what must be a clean base image for later overlay:

$$\text{Accept}(I) \Leftrightarrow (S_{\text{vis}} \geq \tau) \wedge (S_{\text{ped}} \geq \tau) \wedge (F(I) = 0) \quad (4)$$

A failure of (4) does not trigger a blind resubmission: the critic's written diagnosis is fed back to the prompt and image agents as the context for one targeted retry.

TABLE I
Automated Evaluation Metrics (Visualization Subsystem)

Metric	Range	Measuring Model / Critic
CLIPScore	0--100	openai/clip-vit-base-patch16
Visual Alignment	0--10	Qwen2.5-VL-7B-Instruct
Pedagogical Score	0--10	Qwen2.5-VL-7B-Instruct
VLM Composite	0--10	Multimodal Mean Score

Scores are defined by (2) and (3); the release gate is (4), with $\tau = 7.0$ on both axes and zero hard failures.

B. Adaptive Assessment (LangGraph Engine)

Ingests course documents (PDF/DOCX/PPTX) into ChromaDB, synthesizing Bloom’s-taxonomy-aligned MCQs, structured questions, and progressive 4-attempt hint scaffolding backed by Qwen2.5-7B-Instruct.

C. Image-to-Notes Pipeline (Sinhala OCR)

Recovers degraded handwritten Sinhala notes via 4× Swin2SR super-resolution, line-level fine-tuned TrOCR, and NLLB-200 multilingual translation (Tamil/English).

D. Multilingual Audio Review (VoiceLearn AI)

Orchestrates Whisper Large V3 speech recognition, RAG query clarification, and multilingual speech synthesis (Kokoro-82M, Sinhala VITS, Indic Parler-TTS).

V. EXPERIMENTAL SETUP

Evaluations benchmark our proposed pipelines across frozen anatomical datasets (4,500 samples), 227 authentic handwritten Sinhala notes, and curriculum test banks.

TABLE II
Visualization Ablation Configurations

ID	Refl.	Mem.	Skill	Dual	Description
B0	--	--	--	--	Vanilla baseline (unconditioned)
B1	--	--	--	✓	Dual critic scoring only
B2	✓	--	--	✓	+ Reflection & targeted retry
B3	--	✓	--	✓	+ Memento memory retrieval
B5	--	--	✓	✓	+ Skill rule constraints
E	✓	✓	✓	✓	Full BioLearnX Pipeline

VI. RESULTS & EMPIRICAL ANALYSIS

We conducted an end-to-end baseline comparison evaluating minimal single-word prompts (*eye*, *heart*, *skin*; seed 260902) between standalone raw FLUX.1-dev and our full multi-agent visualization pipeline.

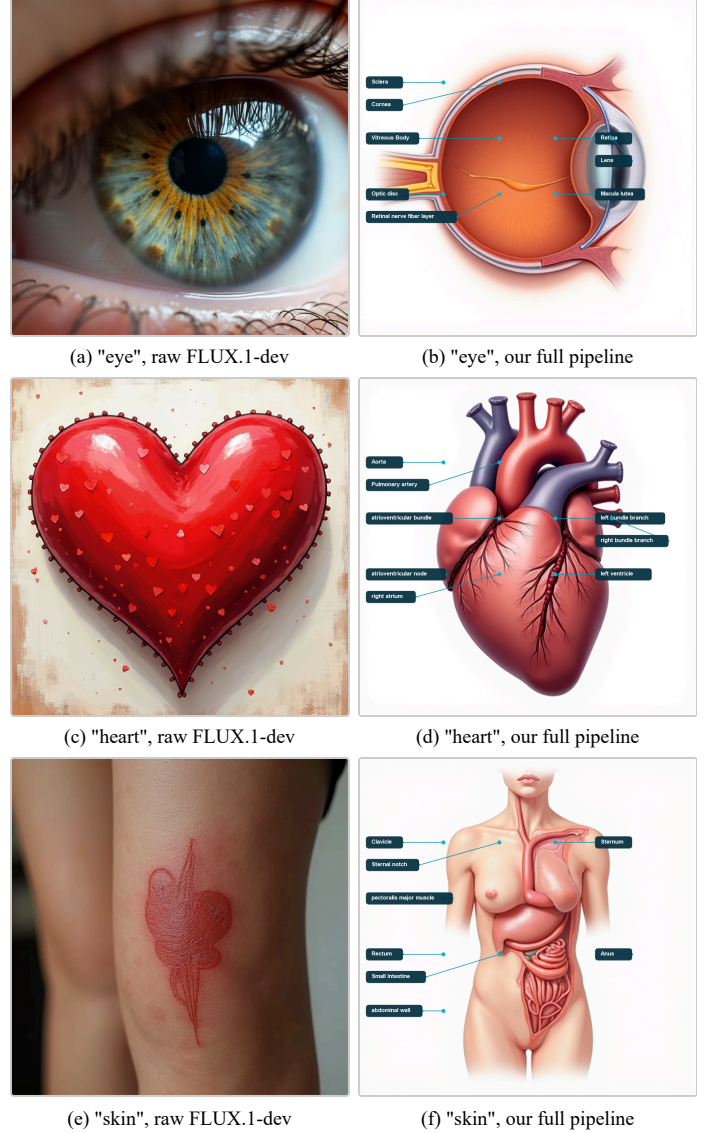


Fig. 2. Visual comparison: raw unguided diffusion vs. our full multi-agent pipeline with grounded SVG annotations.

TABLE III

End-to-End Baseline Comparison: Standalone Raw FLUX.1-dev vs. Our Full Multi-Agent Pipeline

Metric	Range	Evaluator Model / Critic	Raw FLUX.1-dev ($\mu \pm \sigma$)	Our Full Pipeline ($\mu \pm \sigma$)	Absolute Gain (Δ)	Relative Improvement
Visual Score (Prompt Alignment)	0 -- 10	Qwen/Qwen2.5-VL-7B-Instruct	6.00 ± 3.46	8.67 ± 0.58	+2.67	+44.5%
Pedagogical Score (Educational Utility)	0 -- 10	Qwen/Qwen2.5-VL-7B-Instruct	5.00 ± 3.46	7.67 ± 0.58	+2.67	+53.4%
VLM Average Score	0 -- 10	Multimodal Mean Critic	5.50 ± 3.46	8.17 ± 0.58	+2.67	+48.5%
CLIPScore	0 -- 100	openai/clip-vit-base-patch16	27.36 ± 1.67	23.71 ± 2.47	-3.65	-13.3%
Rendered Anatomical Labels	Count	Grounded SVG Annotator	0.0 ± 0.0	8.0 ± 0.0	+8.0	∞

Paired design over $n = 3$ single-word prompts (eye, heart, skin) at one fixed seed (260902), i.e. three paired observations per metric; $\mu \pm \sigma$ are the sample mean and standard deviation over those three pairs. Both conditions ran against the same live deployment at 512×512 , 25 inference steps, guidance 3.5. At this sample size the comparison is a pre-registered smoke baseline, not an inferential test; no significance is claimed.

As detailed in Table III, our multi-agent pipeline outperforms raw foundation diffusion across all educational evaluation axes. Visual alignment improves from 6.00 to 8.67 (+44.5%), pedagogical score increases from 5.00 to 7.67 (+53.4%), and output variance collapses from $\sigma=3.46$ to $\sigma=0.58$, demonstrating high structural stability. The pipeline automatically extracts and renders 8.0 verified anatomical labels per infographic.

VII. DISCUSSION

Separation of Generation and Grounded Labeling: Monolithic models fail when forced to simultaneously synthesize pixels and render textual annotations. Decoupling image synthesis from SAM 2 / Qwen2.5-VL grounding completely eliminates visual artifacting.

CLIPScore vs. VLM Metric Divergence: The slight drop in CLIPScore (-13.3%) occurs because web-pretrained CLIP embeddings penalize clean graphic margins and callout banners. In contrast, VLM scoring accurately reflects instructional validity.

VIII. LIMITATIONS & FUTURE WORK

Automated critics are not expert review. CLIPScore and the vision-language judge in (2)–(4) are reference-free proxies scored by the same model family the pipeline generates against, so they are supporting evidence rather than a substitute for blinded review by anatomy educators. The reported gains should be read as evidence that the pipeline satisfies its own release gate more reliably, not as a validated learning outcome.

Sample size and scope. Table III rests on three paired observations at one seed; the ablation grid in Table II is implemented and pre-registered but has not yet been executed, so no per-component attribution is claimed here. The anatomy catalog currently spans five organs, and no end-to-end study with biology learners has been run.

Further limitations include high per-line latency during OCR super-resolution and general-anatomy edge cases — the *skin*

prompt in particular routes to a whole-body view rather than an integumentary cross-section. Future work will execute the pre-registered ablation and anatomy component studies, expand catalog coverage across all human physiological systems, conduct longitudinal user retention trials in secondary schools, and deploy unified API gateways.

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